

RENAL SYSTEM

NCLEX STUDY GUIDE



Organized for study and printing | 14 content pages

RENAL

Testicular Pathology

1. Testicular torsion = Priority emergency
2. Cryptorchidism
3. Testicular cancer
4. Hydrocele

1. **Testicular torsion** = Priority emergency
 - Life threatening, and is Circulation on the ABC

Assessment

1. Excruciating acute testicular pain.
2. Absent of cremasteric reflex on the affected side.
3. Surgical intervention.

2. Cryptorchidism

- Undescended testicle

Main cause = Prematurity

- No follow up until 6 months
- 80% descend spontaneously.

Surgery after 1 year and if not surgery can cause cancer or infertility

3. Testicular cancer

- Young men (15-30 teen and young adults)
- History off cryptorchism
- Follow up

Assessment

- Painless lump – slow growing
- Gynecomastia
- Increased AFP(α fetoprotein) and Increased HCG Chorionic gonadotropin = Pregnancy test positive
- Dr. Evaluation

4. Hydroceles

- Liquid or fluid in the testicular sac
- Is more frequently in newborn
- Swelling
- Transillumination during examination
- No treatment

Epididymitis

- Acute or chronic inflammation that result of a UTI, STD (chlamydia, gonorrhea), prostatitis.

Assessment

1. Fever with chills
2. Scrotal pain
3. Groin pain
4. Pus and bacterial in the urine (cloudy urine)

Intervention

1. Plenty fluid
2. Antibiotic = Sexual partner too
3. Cold compress in the scrotum.

Kidney Stones

Calcium Oxalate (Alkaline)	Uric Acid
Acid Ash Diet Avoid: ➤ Cocoa - Tea ➤ Cola - Soda ➤ Beans - Rhubarb ➤ Nuts - Tofu ➤ Green Vegetables - Milk and Dairy USE Cranberry Juice	Alkaline Ash Diet Avoid ➤ Organ meats - Bacon ➤ Sardines - Pork ➤ Herrings (pescaditos) - Beef ➤ Salmon - Crab No cranberry juice

Assessment (kidney stones)

1. Excruciating, colicky pain, radiating for groin and genitals
2. Nausea, vomiting, hematuria.

Management

1. Hydration 2500 – 3000 ml/day. (normal = 2500)
2. Opioid analgesics = Morphine
3. Ambulation and Strain the Urine. (colar la orina)

Treatment

- Extracorporeal shock wave lithotripsy (ondas de choque)
- No invasive mechanical procedure for breaking up stones
- Withhold anticoagulants and NSAIDs before.
- Expected after procedure (flank hematoma) = Apply Ice pack

Urinary Incontinence

Stress Incontinence	Urge incontinence	Reflex or overflow Incontinence	Functional Incontinence
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➤ Leak with increased abdominal pressure. ➤ Splinter weakness ➤ More frequently in childbirth or postmenopausal woman (related to decreased estrogens). ➤ Obesity, Multiparity ➤ In men associate with prostate surgery Intervention 1. Kegel exercises - Estrogens - Anticholinergics ➤ Oxybutynin (choice) ➤ Imipramine	➤ Strong urge to void. ➤ Over reactive bladder ➤ Due to stimulants: Alcohol, Caffeine, Smoking, Acidic food Intervention 1. Bladder training Hold 10 minutes, relax and dole void (2 times). 2. Anticholinergics Oxybutynin Imipramine	➤ Neurogenic Bladder (se sale sin sentirlo). ➤ Multiple Sclerosis ➤ Spinal cord injury ➤ Spina bifida Intervention -Crede's method -Pull pubic hair, stroke inner thighs -Open fauces. -Valsava manuever Treatment Betanechol (SQ or oral) (cholinergic) Never IV or IM ➤ Self-catherization at least every 6 hours, less than 500ml of urine in bladder. ➤ (Clean technique)	➤ Mental or physical impairment: Ex: Parkison, Alzheimer. Intervention - Toilet schedule - Easy to remove garnet NEVER BLADDER TRAINING
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Glomerulonephritis (floating RN – pediatric) • Post Streptococcus infection • 2 weeks after: 1. Strep throat - Scarlet Fever 3. Impetigo Assessment Edema around the eyes or facial edema (more prominent in the morning). - Proteinuria = 2g in 24hours Smoky-red Brown or Coca-Cola urine (Hematuria) Hypertension (No. 1 Priority) bc can cause hypertensive encephalopathy = Call Dr if : LOC alteration or confusion Complication - Encephalopathy = ▪ Confusion	Nephrotic Syndrome (floating RN = Pediatric) (Immunocompromised) ➤ Minimal change lession ✓ Autoimmune ✓ Cause unknown Assessment Massive proteinuria >3g/24hours Generalized Edema = weight gain Ascites Scrotal Edema - Decreased urinary output, (frothy urine) or dark yellow Hypercholesterolemia – lipiduria - Lethargy, anorexia, pallor Intervention - Bed rest during acute episode - Diet = restriction of: • Na • K • Cholesterol • water • Normal in protein
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▪ LOC Alteration Call Dr. Immediately ▪ Stroke 2. Renal Insufficiency ○ Increased BUN and Creatinine = Call Dr. 3. Heart failure ○ SOB ○ Crackles Diagnostic • ASO positive = Confirmatory (same in rheumatic fever) Intervention 1. Strict bed rest during Acute phase 2. Diet = restriction of ○ K, Na, water ○ High in calories 3. Medication ○ Diuretic = Furosemide – Bumetanide -Etacrynic acid ○ Antibiotic for streptococcus ○ Antihypertensive med 4. Seizure precaution.	- Never increase or decrease protein, just maintain constant. 3. Steroid (prednisone) = ↓WBC = (risk for infection) Board: Con quien cohort nephrotic syndrome? = Osteomyelitis or Hep A if no diarrhea.
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Pyelonephritis = Urinary Tract Infection (UTI) = Cause No 1 E. coli 80%

Types of UTI

Cystitis	Pyelonephritis
➤ Lower urinary tract infection. ○ Burning urination Treatment ➤ Plenty Fluid ➤ Pyridium (phenazopyridine) ➤ Nitrofurantoin ➤ Cotton panties ➤ Void after sex Side effect (Nitrofurantoin) • Urine roja = expective • Pulmonary congestion • Stop and Call Dr.	➤ Inflammation of the kidney due to bacterial infection -Cause No 1 = E.coli 80% of cases Assessment -Flank pain + Costovertebral tenderness = percussion en kidney = dolor (Note aparte: Renal calculi keyword: pain radiate to the genital or groin area) -Fever with chills -Nausea and vomiting -Malaise, fatigue -Dysuria, burning urination -Cloudy, bloody and foul-smelling urine -Elderly= confusion, LOC alteration = hacerle analysis de orina bc Cause no 1 de delirium in elderly is UTI

	<u>Diagnostic</u> 1. Clean catch method = 1st voiding in the morning 2. Urinary Culture = Confirmatory = Mid-stream -Leukocyte esterase positive =UTI + 100,000 col x field = UTI <u>Intervention (pyelonephritis)</u> 1. Antibiotic a. Sulfa (TSM) S/E = Crystalluria, (plenty Fluids) and Skin rash bc Steven Johnson b. Ciprofloxacin S/E = Achilles tendon rupture = stop if heel pain. Note: calf pain = (Statins)=rhabdomyolysis c. Plenty fluid <u>Prevention:</u> -Encourage to void every 2 hours -Shower rather than bath -Cleanse from front to back. -Void immediately after sexual intercourse -Wear cotton undergarment and loose clothing.
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Prostatitis

- Inflammation of the prostate gland is commonly caused by infection or not.

Causes

1. E-Coli
2. STI = Chlamydia (Tx: azithromycin) or Gonorrhea
3. Viral illness
4. Decreased sexual activity.

Assessment

1. Fever + Chills
2. Dysuria (dificultad para orinar)
3. Low back pain
4. Urethral discharge (increase with prostate palpation)
5. DRE Digital rectal Exam= prostate gland=
 - Bulging and warm to touch

Intervention and Treatment

1. Plenty fluid
2. Use tub or sitz bath
3. Administer antibiotic, stool softener
4. No sexual intercourse, masturbation or prostatitis massage (**Only Preventive**)
5. Avoid
 - Spicy food
 - Alcohol

- prolonged automobile rides

Benign Prostatic Hypertrophy (BPH)

- Is a slow enlargement of the prostate gland with hypertrophy and hyperplasia of normal tissue
- Resulting in partial or complete obstruction of urine
- More frequently after 50y/o

Assessment (BHP)

- Early: Nocturia (frequently urine during night)
 - Decreased Size and force of urinary stream.
 - Inability to start (hesitancy) or continue the urinary stream.
- Post void dribbling (goteo)
- Hematuria
- Feeling of incomplete empty bladder
- Bladder distention
- Frequently UTIs

Treatment

1. Encourage fluid intake 2500-3000 ml/day
2. Prepare for urinary catheterization in case of obstruction.
3. Medication
 - a. 5 α Reductase inhibitors
 - Finasteride
 - Dutasteride

S/E

- ↓ Libido + impotence
- Orthostatic hypotension
- Pregnant woman = Avoid contact with tablets or use gloves to prevent teratogenesis)
- b. α Blockers = for HTN
 - Tamsulosin
 - Terazosin

S/E

- Orthostatic hypotension
- Drowsiness

- c. Saw Palmetto (herb) just for BPH

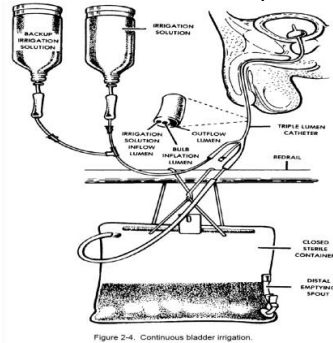
4. Surgical Intervention
 - TURP = (La prostata por la uretra)
 - Retropubic/Suprapubic resection (through bladder) (abdomen, arriba)

Transurethral resection prostate (TURP)

- Teaching Pt that is expected bladder spasm after surgery due to clots.
- **Board: If pain or increased spasm what to do?**
- ✓ Intervention: Flushing catheter with 10 ml of NaCl

All Pt with this surgery After procedure:

✓ CBI catheter: Triple lumen catheter



- Cath secured very snugly on inner thigh to prevent kink or obstruction
- During the first 24h there is expective:
 - ✓ pinkish drainage with small to moderate clots.
 - ✓ If bright red or big cloth= Call Dr.
- Expective burning urination during first week after surgery
- Remove catheter 2-5 days after.

Complication

1. Hemorrhage.
2. **TURP Syndrome**
 - ↑Fluid overload = Delusional hypothermia too **low Na.**
3. Brain Edema (same to SIADH)
 - LOC Alteration
 - Confusion
 - Seizure.

Call Dr

Renal Insufficiency Acute or Chronic

Function = filtration

Acute:

- Rapid loss of kidney function from renal cells Damage.
- Occur abruptly, can be reversible
- Glomerulo Filtration Rate (GFR)= less than 90 (normal = 90 – 120)

Chronic kidney injury: mantenido less than 60 per 3 months or more

Causes

Pre- Renal (above kidney)	Intra-Renal (direct damage to kidney)	Post-Renal (obstruction) (SATA) del kidney para abajo.
➤ Sudden or severe ↓BP ➤ Dehydration ➤ Hypovolemic shock ➤ Hear attack ➤ CHF ➤ Sepsis (septic shock) ➤ Liver Failure ➤ Anaphylactic reaction	➤ Glomerulonephritis ➤ Bleeding in the kidney ➤ LES ➤ Pyelonephritis (chronic) ➤ Chemotherapy ➤ Dyes ➤ Metformin	➤ Renal calculi ➤ Cervical cancer ➤ Colon cancer ➤ Endometrial cancer ➤ Prostate cancer ➤ Kidney stone ➤ Clots in urine tract ➤ Neurogenic bladder

➤ Severe Burn = hypovolemic shock ➤ Renal Artery Stenosis ➤ Atherosclerosis ➤ Blood pressure medication ✓ NSAID ✓ ASA	➤ Zoledronic Acid (Osteoporosis IV) S/E Atrial fib/ renal insufficiency ➤ Multiple myeloma ➤ Scleroderma ➤ Purpura ➤ Alcohol ➤ Heavy metals ➤ Cocaine	➤ BPH
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Phases of Acute Kidney injury

1. Oliguric phase = poca orina
2. Diuretic or polyuric = mucha orine
3. Recovery

Note: anuria = no orina

1. Oliguric Phase (poca orina) (fluid retention)

- Decreased Urinary Output (normal 1500 ml/24h)
 - Less than 400 ml/day
 - Lasting 8-15 days

Assessment

1. Signs of fluid overload
 - SOB
 - Lung crackles
 - Tachycardia with bounding pulse
 - Increased Pulmonary Edema (pink frothy sputum)
2. Signs of Uremia
 - Anorexia, nausea, vomiting
 - Pruritus
 - LOC alteration, confusion, drowsiness, paresthesia
 - Uremic encephalopathy

Nota: Como yo se que esta saliendo de la phase oliguric? = Mejora el LOC, mas atento, menos confundido.

3. Metabolic Acidosis (K ↑)
 - Kussmaul respiration

Labs

1. ↑ BUN, ↑ Creatinine
2. K ↑
3. Fluid retention
4. PO₄ ↑
5. GFR ↓
6. Ca ↓
7. Na (normal or low)

Treatment

1. Diuretic (furosemide)
2. Fluid restriction
3. Hemodialysis

2. Phase Diuretic or polyuric

- Urine output rise slowly followed by diuresis (4-5 L/day).

Assessment

1. Dehydration (thirst)
2. Hypovolemia
3. Hypotension + tachycardia with weak pulse.
4. LOC Improves

Labs

1. Gradual decreased BUN and Creatinine
2. K ↓
3. Na ↓
4. Hypovolemic
5. GFR ↓ = No good

Treatment

1. IV fluids.

3. Recovery Phase = Lasting months to years

- Increased GFR
- Stabilization of BUN, creatinine
- Improve of strength
- Improve of memory

Chronic Kidney Disease

Oligúrica mantenida = Irreversible

- Is a slow, progressive irreversible loss of kidney function with GFR less than 60 ml/min for 3 months or more.

Causes (CKD)

1. May follow Acute Kidney Injury (AKI).
2. Diabetes
3. HTN
4. Metabolic Disorders
5. Chronic Urinary obstruction
6. LES = Autoimmune disorder

Assessment

1. Cardiovascular
 - Hypertension
 - Heart failure
 - Edema
 - Pericardial effusion (cardiac tamponade)
 - Pericarditis (pericardial friction rub).
2. Respiratory
 - CHF
 - SOB, lung crackles
 - Kussmaul respiration (bc metabolic acidosis)
 - Uremic Fetor (respiration huele a orina)
 - Halitosis (mal aliento)
3. Hematological Manifestation
 - Anemia 9eritropoitin es el tx : epoetin alpha starting when less than 10.5 and stop in 11-12 to prevent clots)

- Bruising
- Abnormal Bleeding
- 4. GI Manifestation
 - Anorexia, nausea, vomiting
 - Constipation diarrhea
 - Stomatitis
 - Metallic taste
 - Uremic gastritis
- 5. Integumentary
 - Dry Skin
 - Pruritus (uremic frost)
 - Yellow gray pallor
- 6. Musculo skeletal
 - Pathologic fracture
 - Renal Osteodystrophy (calcio bajo)
- 7. Reproductive
 - Impotency
 - ↓ libido, ↓ Fertility, ↓ or absent menstruation.
- 8. Neurologic
 - Uremic encephalopathy
 - Asterixis (Intentional hands tremors) (cirrhosis =writing difficulty)
 - Ataxia

Nota: dysarthria = difficulty to speech = speech therapy)

- Inability to concetrate
- Lethargy
- Clonus (before seizure)
- Slurred speech
- Coma.

Intervention (renal patients)

1. Diet= **Restriction of:**
 - Na, K, fluids, PO4, Protein
 - Acid restriction
 - Protein restriction in all except peritoneal Dyalysis (have to ↑ protein bc Ascitis).
2. Oral care to prevent stomatitis
3. Skin care to prevent pruritus (Benadryl)
4. Daily weight
5. Diuretic
 - Furosemide
 - Bumetanide (board)
6. Anemia
 - Epoetin $\propto$
 - Staring when HB less than 10 – 10.5.
 - Stop when HB = 11-12 to prevent thrombosis
 - Acid folic
 - Vit B12
7. Hyperphosphatemia
 - PO4 binding
 - Sevelamer
 - Aluminum Hydroxide

With food
or

- Anphogel snack

Nota: Phosphate binding (sevelamer, aluminum hydroxide, anphogel) para el kidney se usa como bindin with food or snack. For GI= lejos de las comidas.

8. Hyperkalemia

- Sodium polystyrene (keyaxalate) oral o enema.
- Side Effect = Diarrhea =Stop + Call Dr.

Hemodialysis

- Clean technique
- Is the process of cleansing the patient's blood by diffusion, osmosis, ultrafiltration.
- 3 times per weeks

Function

- Cleanses the blood of wastes products
- Remove urea, creatinine, uric acid.
- Remove excess of body fluids.
- Correct electrolytes levels in the body

Dialysate bath (water)

- Water or electrolytes
- Need not to be sterile

Access for Hemodialysis

1. Subclavian and femoral catheter.

- Short term or temporary 4 – 6 weeks

Nota: hasta que la fistula este ready.

➤ Femoral

- More risk for infection
- Patient should not sit up more than 45°

2. External Arteriovenous shunt (fistula)

- Permanent access of choice for client with CKD.
- The fistula is created surgically by anastomosis of the large vein with artery in the arm.
- Maturity 4-6 weeks – Hand exercises = Squeezing the ball
- Before use, check for patency
- Bruits to auscultation
- Thrill to palpation

Fistula Complication

- Arterial steal Syndrome
 - Hand circulation compromise
 - Pallor, coldness, cyanosis = Call Dr.

3. Internal Arteriovenous graft

- Anastomosis of an artery to a vein using artificial graft (Puerco).
- Can be used 2-4 weeks
- Allow freedom movement
- More risk for:

- Clots
- Aneurism
- Infection

Nota: se usa cuando la otra fistula no dio resultado

Interventions (general hemodiálisis)

1. Monitor vital signs before, during and after = specially BP
- No.1 Complication = Hypotension
- Temperature = expected mild increased of temperature during hemodialysis 100.3 – 100.4 F
2. Weight client before and after.
3. Check for bleeding (bc dialysate contained heparin)
4. Adequate nutrition = Client may eat before and during dialysis.

Withhold Medication before

- Antihypertensive medication
- Antibiotic
- Aminoglycosides
- Cephalosporin
- Anti TB
- Cimetidine (no se mezcla con nadie)
- Aspirin
- Digoxin
- Water soluble vitamins: B1, B3, B6, B12, Ascorbic Acid (vit C), Folic Acid.
- Antiviral
- Allopurinol (gota)
- Fluconazole

Complications Hemodialysis

1. Hypotension
2. Disequilibrium syndrome (ocurre en los primeros procesos)
 - Brain Edema
 - Occurs during 1st hemodialysis process in children or elderly.
 - Self-limits.

Assessment

1. LOC alteration
2. Seizure
3. Headache
4. Increased BP
5. Nausea and vomiting
6. Muscle cramp.

Treatment

1. Mannitol
2. IV fluid (hypertonic solution)
3. Albumins
4. Seizure Precaution

Peritoneal Dialysis (sterile technique)

- Exchange of wastes fluids and electrolytes
- Occur in the peritoneal cavity

- Need more time than hemodialysis
- Daily.

Continuous Ambulatory Peritoneal Dialysis

- Does not required machine for the procedure.
- Promote patient independence.

Procedure

- A siliconized catheter is surgically placed into the abdominal cavity
- 3-5 cm below umbilicus.

PHASES

Infusion	Dwell	Outflow
1-2 liters of dialysate via gravity into peritoneal cavity (10-20minutes)	Time in abdominal cavity. 30 minutes- 8 hours Collection bag up Above catheter level	Collection bag down. Drainage of wastes products.

Dialysate additives (if needed)

- Heparin
- Potassium, (no en la misma bolsa con el antibiotic)
- Antibiotic
- Insulin = Expected increased blood glucose

Complication of peritoneal Dialysis

1. Peritonitis (infection)
 - Cloudy outflow
 - If cloudy or opaque outflow =stop and call Dr. (And send to lab la muestra)
2. If blood drainage = check for period (women).
 - If not period = Call Dr.
3. If stop outflow
 - Kink or obstruction
 - Change position (side to side)
4. If decreased outflow = Constipation
 - Increased fiber intake (board)
 - Stool softener.
5. Abdominal pain or cramps
 - Decreased rate
 - Warm dialysate
6. Leakage around catheter = (expected)
 - Change dressing = bc is expected during 1st week

Nota: el liquid que sale tiene que ser clear, todo los cloudy or dark lleva follow up.

Renal Transplant. (Justice principle)

- After procedure is expected mild hematuria.
- Expected urinary output = >100ml/h during 24-48 hours.
- Replace lost volume by ml.
- If urine output = 50 ml/h = Call the Dr.
- Life long neutropenic precaution
 - Isolate bc immunocompromise
 - Cohort only with healthy

Rejection	
Hyperacute	Acute
➤ During 1st hours to days (24-48hours). ✓ Wrong matching ✓ Fever ✓ Tenderness (flank pain) ✓ ↓ Urine flow or stop Intervention ➤ Immediately removal of kidney.	➤ Days to weeks ✓ ↓ Urinary output ✓ ↑ weight or edema. Intervention ➤ Increase dose of immunosuppressant medication.
Chronic	
➤ Months to years ✓ ↑ BUN ✓ ↑ creatinine ✓ ↓ GFR Intervention ➤ New transplant ➤ Hemodialysis	
P-a ra My T- transplant C- curar	Treatment Prednisona Mycophenolate. Tacroymus Cyclosporine } Risk for infection